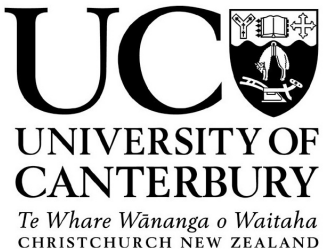
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Energy and Momentum in General Relativity:

Gravitational Waves in Asymptotically Flat Space-Times

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Abstract

The difficulty in defining energy and momentum in general relativity is described, and asymptotic space-times are introduced. We describe two regimes, the Bondi-Sachs formalism, built around null hyper-surfaces, and the Spin-coefficient formalism using the so called spin-coefficients, described most elegantly using the 2-spinor calculus. We find how these two regimes are connected and describe the connection between certain key definitions. In particular, these regimes each allow a definition of mass and mass loss that relates to the energy in gravitational waves, and we describe their equivalence.

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Chapter 1

Introduction

In general relativity, there is not a well-defined universally applicable definition of gravitational energy. The standard local description of energy for non-gravitational energy in terms of the energy-momentum tensor breaks down for gravitational energy, as gravitational energy is inherently non-local. One would imagine the gravitational energy would have to be proportional to the curvature of the space-time, but locally we find that the curvature disappears due to the equivalence principle. Thus any tensorial quantity defined from the curvature is also zero locally. This highlights the idea that gravitational energy is a global property, or at least non-local. In this report, we will investigate *asymptotically flat* spacetimes and look at the connection in definitions between the *Bondi-Sachs* formalism, and the *Spin-coefficient* formalism with the particular focus on gravitational waves and energy. These systems are essentially built off the ideas of Bondi [1]. Each of them are well suited to the study of gravitational waves, and we can choose the coordinate system for both to be essentially the same. This coordinate system is based on a family of geodesics, labelled by u that are parametrised by a coordinate r or $\tilde{r}$, and at each point in the two-parameter system we describe the remaining coordinate freedom with a 2-sphere structure. In the formulations as given, the coordinates only differ by the parametrising coordinate and the transformation between definitions with different parameters is explored. The formalisms developed will be used to solve for solutions in vacuum space that are asymptotically flat, and will model the space around isolated sources. The dynamic mass of the sources will be understood as the *Bondi mass* and will be defined such that there is agreement with the standard Schwarzchild mass for the static case. Between static periods, mass loss of the source due to the outwardly radiated gravitational wave energy will be understood and given by the *Bondi mass loss* formula. The meaning of initial data and boundary conditions will be considered, and differences in how they are prescribed between formalisms will be appreciated.

Understanding how gravitational waves relate to energy and momentum is fundamentally important for making sense of the physical observations of gravitational waves. The kinds of gravitational waves that we are able to detect can be modelled as sourced by isolated systems as an approximation, thus giving the asymptotically flat spacetime real world significance. The modern day interpretation of asymptotic flatness in terms of conformal compactification is mentioned, and future work is suggested.

Chapter 2

Bondi-Sachs Formalism

First described in the 1960s by Bondi [1], the Bondi-Sachs formalism is built around a foliation of space-time via null, outgoing hypersurfaces. We begin by defining a a coordinate system (x^a), $a \in \{0,1,2,3\}$ that encodes the structure of outgoing null rays, thus making them easy to study. Take $x^0 = u$, to satisfy $g^{ab}(\partial_a u)(\partial_b u) = 0$, then the hypersurfaces defined by $u = const$ are null, that is their conormals, $l_a = \partial_a u$, are null. We then see that the normal direction vector $l^a = g^{ab} l_b$ has the curves tangent to it, known as 'rays' in analogy with the propagation of light, that are null geodesics. Assume there is some parametrisation along these rays $x^1 = r$, then at each (u, r) the remaining freedom can be described as a 2-sphere that can be described by two angular coordinates $x^{\mathcal{A}}$, where $\mathcal{A}, \mathcal{B}, \mathcal{C}, \dots \in \{2,3\}$, constant along null rays. We can then choose r to be the parameter such that it also satisfies $\det(g_{\mathcal{AB}}) = r^4 q$, where $q = q(x^{\mathcal{A}})$ is the determinant of the unit sphere metric $q_{\mathcal{AB}}$. While defining the null hypersurfaces that foliate spacetime the u coordinate also acts as a retarded time coordinate with the usual interpretation of $u = t - r$. These conditions mean that $g^{uu} = 0$, $g^{u\mathcal{A}} = 0$, which imply $g_{rr} = 0$, $g_{r\mathcal{A}} = 0$. We can then write any metric satisfying these conditions in the Bondi-Sachs form, given as

$$g_{ab} = \begin{bmatrix} -\frac{V}{r}e^{2\beta} + U^{\mathcal{A}}U^{\mathcal{B}}r^2 h_{\mathcal{AB}} & -e^{2\beta} & -r^2 h_{\mathcal{AB}}U^{\mathcal{B}} \\ -e^{2\beta} & 0 & 0_{\mathcal{A}} \\ -r^2 h_{\mathcal{AB}}U^{\mathcal{A}} & 0_B & r^2 h_{\mathcal{AB}} \end{bmatrix},$$

with inverse

$$g^{ab} = \begin{bmatrix} 0 & -e^{-2\beta} & 0^{\mathcal{A}} \\ -e^{-2\beta} & \frac{V}{r}e^{-2\beta} & -U^{\mathcal{A}}e^{-2\beta} \\ 0^B & -U^B e^{-2\beta} & \frac{1}{r^2}h^{\mathcal{AB}} \end{bmatrix},$$

where $0^{\mathcal{A}} = (0,0)$. The remaining degrees of freedom in the metric manifest as functions of the coordinates, V, β, $U^{\mathcal{A}}$, $h_{\mathcal{AB}}$, and $g_{\mathcal{AB}} \coloneqq r^2 h_{\mathcal{AB}}$. From the determinant condition defining r that relates $g_{\mathcal{AB}}$ and $h_{\mathcal{AB}}$, we can see $\det(h_{\mathcal{AB}}) = q(x^{\mathcal{A}})$. As $h_{\mathcal{AB}}$ must also be symmetric, this allows us to fix $h_{\mathcal{AB}}$ with two more functions of the coordinates γ, and δ. We give the general form as in Winicour [2]:

$$h_{\mathcal{AB}} = \begin{bmatrix} \cosh(2\delta)e^{2\gamma} & \sin\theta\sinh(2\delta) \\ \sin\theta\sinh(2\delta) & \sin^2\theta\cosh(2\delta)e^{-2\gamma} \end{bmatrix},$$

here $h_{\mathcal{AB}}$ acts as the the 2-sphere metric for coordinates $x^{\mathcal{A}}$.

2.1 Einstein equations

The Einstein equations are $E_{ab} := R_{ab} - \frac{1}{2}g_{ab}R - 8\pi T_{ab} = 0$, for Ricci tensor, R_{ab}, Ricci scalar, R, and energy-momentum tensor T_{ab}, expressed in units where $c = G = 1$. We have also assumed that the cosmological constant is vanishing, $\lambda = 0$. Where $\nabla_a T^a_b = 0$, we have the divergence free condition $\nabla_a E^a_b = 0$, implied by the Bianchi identities of the Riemann tensor[1]. Here, ∇_a is the unique covariant derivative operator (Levi-Civita) that is both torsion free, and metric preserving. Then, we find that only the *main equations*;

$$E^u_a = 0, \tag{2.1}$$

$$E_{\mathcal{AB}} - \frac{1}{2}g_{\mathcal{AB}}g^{\mathcal{CD}}E_{\mathcal{CD}} = 0\,, \tag{2.2}$$

and *supplementary equations*:

$$\partial_r(r^2e^{2\beta}E^r_u) = 0,\ \partial_r(r^2e^{2\beta}E^r_{\mathcal{A}}) = 0\,,$$

are required to resolve the complete solutions to the Einstein equations. In order to show this, first note that (2.1) is equivalent to $E_{ra} = 0$. We can then assume the main equations hold, and evaluate the divergence free condition in the coordinate system set up. The $b = r$ component reduces to $g^{\mathcal{AB}}E_{\mathcal{AB}} = 0$, so that the 2.2 can be seen to be equivalent to $E_{AB} = 0$, while the $b = u, A$ components reduce to the supplementary equations. We are left to show that $E_{uu} = 0 = E_{uA}$, or correspondingly $E^r_u = 0 = E^r_{\mathcal{A}}$, in order to have complete equivalence. The supplementary equations give this final piece as they show $E^r_u, E^r_{\mathcal{A}}$ are identically zero if they are zero on some finite worldtube, or the limit as $r \to \infty$. ‘

Designating equations (2.1) as the *hypersurface equations*, and equations (2.2) as the *evolution equations*, in terms of the unknown functions of the metric the hypersurface equations are found:

$$\partial_r\beta = \frac{r}{16}h^{\mathcal{AC}}h^{\mathcal{BD}}(\partial_r h_{\mathcal{AB}})(\partial_r h_{\mathcal{CD}}) + 2\pi r T_{rr},$$

$$\partial_r\left(r^4e^{-2\beta}h_{\mathcal{AB}}\left(\partial_r U^{\mathcal{B}}\right)\right) = 2r^4\partial_r\left(\frac{1}{r^2}D_{\mathcal{A}}\beta\right) - r^2h^{\mathcal{CD}}D_{\mathcal{C}}(\partial_r h_{\mathcal{AD}}) + 16\pi r^2 T_{r\mathcal{A}},$$

$$2e^{-2\beta}\partial_r V = \mathcal{R} - 2h^{\mathcal{AB}}\left[D_{\mathcal{A}}D_{\mathcal{B}}\beta + (D_{\mathcal{A}}\beta)\,(D_{\mathcal{B}}\beta)\right] + \frac{e^{-2\beta}}{r^2}D_{\mathcal{A}}\left[\partial_r\left(r^4U^{\mathcal{A}}\right)\right]$$
$$-1/2r^4e^{-4\beta}h_{\mathcal{AB}}(\partial_r U^{\mathcal{A}})(\partial_r U^B) + 8\pi\left(h^{\mathcal{AB}}T_{\mathcal{AB}} - r^2T\right),$$

where $D_{\mathcal{A}}$ and $\mathcal{R}$ are the (Levi-Civita) covariant derivative and Ricci scalar with respect to $h_{\mathcal{AB}}$, while $T = T^a_a$ is the trace of the energy-momentum tensor with respect to the metric. The evolution equations then most easily follow from an introduced complex dyad as in Winicour [2]; Where $(M^a, \bar{M}^a)$ denotes the dyad we take it to be tangent to null hypersurfaces, $M^a l_a = 0$, and normalised such that $h^{\mathcal{AB}} = M^{\mathcal{A}}\bar{M}^{\mathcal{B}} + M^{\mathcal{B}}\bar{M}^{\mathcal{A}}$. Given that $h^{\mathcal{AB}}$ is used to lower and raise 2-tensors on the sphere we find $M_{\mathcal{A}}\bar{M}^{\mathcal{A}} = 1$, while $M_{\mathcal{A}}M^{\mathcal{A}} = 0$. In our coordinates these conditions give the dyad with components $M^a = (0, 0, M^{\mathcal{A}})$ unique up to a phase, which can be fixed as necessary by additional constraints. The evolution

[1]Note we write the tensor indices with no indent where the tensor in question is symmetric as there is no ambiguity.

equations are then equivalent to the single complex equation $M^{\mathcal{A}}M^{\mathcal{B}}E_{\mathcal{AB}} = 0$, which is written in terms of the unknown functions:

$$m^{\mathcal{A}}m^{\mathcal{B}}\Bigg[r\partial_r(r\partial_u h_{\mathcal{AB}}) - \frac{1}{2}\partial_r(rV\partial_r h_{\mathcal{AB}}) - 2e^{\beta}D_{\mathcal{A}}D_{\mathcal{B}}e^{\beta} + h_{\mathcal{AC}}D_{\mathcal{B}}\left[\partial_r\left(r^2U^{\mathcal{C}}\right)\right] - \frac{1}{2}r^4e^{-2\beta}h_{\mathcal{AC}}h_{\mathcal{BD}}\left(\partial_r U^{\mathcal{C}}\right)\left(\partial_r U^{\mathcal{D}}\right) + \frac{r^2}{2}\left(\partial_r h_{\mathcal{AB}}\right)\left(D_{\mathcal{C}}D^{\mathcal{C}}\right) + r^2U^{\mathcal{C}}D_{\mathcal{C}}\left(\partial_r h_{\mathcal{AB}}\right) - r^2\left(\partial_r h_{\mathcal{AC}}\right)h_{\mathcal{BD}}\left(D^{\mathcal{C}}U^{\mathcal{D}} - D^{\mathcal{D}}U^{\mathcal{C}}\right) - 8\pi e^{2\beta}T_{\mathcal{AB}}\Bigg] = 0\,.$$

These comprise a system of differential equations that can be integrated to find a solution for each of the unknown functions at initial u via the hypersurface equations, and their time evolution via the evolution equation.

2.2 Solution in asymptotically flat case

We will find it sufficient to define an asymptotically flat spacetime as one where the Riemann curvature becomes negligible at large distances from some region of the manifold, in the sense that it reduces to Minkowski spacetime at these large distances. In order to ensure this, we will assume that the metric reduces to the Minkowski metric (with retarded time coordinate u) for large r, stated as conditions:

$$\lim_{r\to\infty}\beta = \lim_{r\to\infty}U^{\mathcal{A}} = 0\,, \quad \lim_{r\to\infty}\frac{V}{r} = 1\,, \quad \lim_{r\to\infty}h_{\mathcal{AB}} = q_{\mathcal{AB}}\,.$$

We will also assume that each of the functions have an expansion in powers of $1/r$, this is sufficient for the above limits to hold but it is to be noted that this condition is not necessary. It is shown in Winicour [3] that logarithmic terms can enter into the expressions, in which case the *peeling property* of the Weyl tensor that will be shown to result in Chapter 3 does not hold in general. Finally, we will assume that the vacuum Einstein equations hold, that is $T_{ab} = 0$.

The Einstein equations as stated in Section 2.1, with the key assumptions applied, have a unique solution only when some additional information is supplied. One method is to supply the following data:

- The 2-metric, $h_{\mathcal{AB}}(u_0, r, x^{\mathcal{A}}) = q_{\mathcal{AB}} + \frac{c_{\mathcal{AB}}(u_0,x^{\mathcal{C}})}{r} + \ldots$, on an initial hypersurface defined by $u = u_0$.
- The first coefficient for the 2-metric, $c_{\mathcal{AB}}(u, x^{\mathcal{C}})$, for some time period $u \in (u_0, u_1)$.
- The function $M(u_0, x^{\mathcal{A}}) \coloneqq -\frac{1}{2}\lim_{r\to\infty}\left[V(u_0, r, x^{\mathcal{B}}) - r\right]$, for u_0, known as the *mass aspect*.
- The function $L_{\mathcal{A}}(u_0, x^{\mathcal{C}}) \coloneqq -\frac{1}{6}\lim_{r\to\infty}\left(r^4e^{-2\beta}h_{\mathcal{AB}}\partial_r U^{\mathcal{B}} - r\eth^{\mathcal{B}}c_{\mathcal{AB}}\right)$, for u_0, known as the *angular momentum aspect*.

Here, $\eth_{\mathcal{A}}$ is the (Levi-Civita) covariant derivative with respect to $q_{\mathcal{AB}}$. With these the hypersurface equations can be integrated for the asymptotic solutions of each variable at u_0. For example, we find that

$$\beta(u_0, r, x^{\mathcal{A}}) = -\frac{1}{32r^2} c^{\mathcal{AB}} c_{\mathcal{AB}} + O(r^{-3}) \,,$$

where Big O notation is used in the standard sense, for $r \to \infty$. Each of the asymptotic variables can then be determined through u via the evolution equation and supplementary conditions, using the leading order terms. In particular, we find that the evolution of the mass aspect is governed by the E^r_u supplementary equation, and reduces to

$$\partial_u M = \frac{1}{2} \eth_{\mathcal{A}} \eth_{\mathcal{B}} N^{\mathcal{AB}} - \frac{1}{2} N_{\mathcal{AB}} N^{\mathcal{AB}} \,. \tag{2.3}$$

The evolution of the angular momentum aspect is determined similarly from the $E^r_{\mathcal{A}}$ supplementary condition

$$-3\partial_u L_{\mathcal{A}} = \eth_{\mathcal{A}} M - \frac{1}{4} \eth^{\mathcal{B}} \left(\eth_{\mathcal{B}} \eth^{\mathcal{C}} c_{\mathcal{AC}} - \eth_{\mathcal{A}} \eth^{\mathcal{C}} c_{\mathcal{BC}} \right) + \frac{1}{8} \eth_{\mathcal{A}} \left(c_{\mathcal{BC}} N^{\mathcal{BC}} \right) - \eth_{\mathcal{B}} \left(c^{\mathcal{BC}} N_{\mathcal{CA}} \right) + \frac{1}{2} c^{\mathcal{BC}} \eth_{\mathcal{A}} N_{\mathcal{BC}} \,, \tag{2.4}$$

where the *news tensor* is defined by

$$N_{\mathcal{AB}} \coloneqq \partial_u c_{\mathcal{AB}} \,, \tag{2.5}$$

2.3 Shear, news and Bondi mass

We can define a function

$$\sigma^0 \coloneqq \lim_{r \to \infty} \left(M^{\mathcal{A}} M^{\mathcal{B}} c_{\mathcal{AB}} \right) \,,$$

stated in terms of a dyad $Q^{\mathcal{A}} = \lim_{r \to \infty} M^{\mathcal{A}}$ as $\sigma^0 = Q^{\mathcal{A}} Q^{\mathcal{B}} c_{\mathcal{AB}}$. This is equivalent to the asymptotic shear of the geodesic foliation determined by u, in analogy with optics[2], with equivalent expression

$$\sigma^0 = \lim_{r \to \infty} \left(M^a M^b \nabla_a \nabla_b u \right) \,, \tag{2.6}$$

given via Jordan, Ehlers, and Sachs [4]. From this we define the *news function*:

$$N \coloneqq Q^{\mathcal{A}} Q^{\mathcal{B}} N_{\mathcal{AB}} \,,$$

and we can see that $\partial_u \sigma^0 = N$. When we define the *Bondi mass*,

$$m(u) \coloneqq \oint M dS \,,$$

integrated over the 2-sphere, (2.3) gives the *Bondi mass loss*

$$\dot{m}(u) = \oint \partial_u M \, dS^2 = -\frac{1}{2} \oint N_{\mathcal{AB}} N^{\mathcal{AB}} \, dS^2 = - \oint |N|^2 \, dS \,, \tag{2.7}$$

[2]We have used the same symbol here as in Chapter 3, this will be justified in Chapter 4.

where the second equality is given by the fact that $\eth_{\mathcal{A}}\eth_{\mathcal{B}}N^{\mathcal{AB}}$ integrates out, while the third equality is given by the determinant condition $\det(h_{\mathcal{AB}}) = q$. This demonstrates the result that the Bondi mass can only decrease, and its change over time is entirely specified by the news function. Via (2.3) and (2.4), we see the time evolution of the entire system is determined by $c_{\mathcal{AB}}$ which has two real free parameters. Alternatively we can specify the complex asymptotic shear, σ^0, or the complex news function, N, as one free complex parameter can hold the information in both free parameters of $c_{\mathcal{AB}}$. We further justify this as a reasonable definition of mass, by looking at the static case. If the system is independent of u, which is equivalent to saying that $N = 0$ or $N_{\mathcal{AB}} = 0$, then m is a constant. In this case, the constant m matches with the standard mass, which can be seen in the spherically symmetric case where the metric reduces to the Schwarzschild metric, and m is the Schwarzchild mass. Collecting these ideas, outgoing gravitational waves are characterised by the a non-zero news, and non-zero news implies that there is a mass loss of the source. We can interpret the mass loss as being converted into gravitational wave energy, and this allows us to quantify the energy of the gravitational waves. It is possible to understand the angular momentum in a similar way through the integration of $L_{\mathcal{A}}$ and using (2.4).

Chapter 3

Spin-coefficient (Newman-Penrose) formalism

The Newman-Penrose formalism is based around null tetrads that can be equivalently expressed in terms of 2-spinors. The formulation of spinors used, the 2-spinors, is as in Penrose and Rindler [5] and its essential components are discussed in Appendices A.2 and A.3. It reformulates the Einstein field equations into a set of complex first order differential equations in terms of the (complex) scalar components of each of the tetrad covariant derivatives. In the following section we will adopt the abstract index approach as introduced by Penrose [5], described in Abstract A.1. Lowercase Latin indices $(a, b, \dots)$ denote abstract tensors, uppercase Latin indices $(A, B, \dots, A', B', \dots)$ denote abstract spinors, and bold-face indices $(\boldsymbol{A}, \boldsymbol{B}, \dots, \boldsymbol{A'}, \boldsymbol{B'}, \dots, \boldsymbol{a}, \boldsymbol{b}, \dots)$ are used for the numbers 0 to 3 with the standard Einstein summation convention, denoting the relevant components of the tensors and spinors. We begin by setting up a null tetrad[1], $(l^a, n^a, m^a, \bar{m}^a)$, that satisfies the following additional relations $l^a m_a = n^a m_a = l^a \bar{m}_a = n^a \bar{m}_a = 0$ and $l^a n_a = -m^a \bar{m}_a = 1$. Using 2-spinors (hereafter: spinors) we set up a spin frame, (o^A, ι^A), to equivalently define the tetrad with the definitions $l^a = o^A o^{A'},\ n^a = \iota^A \iota^{A'},\ m^a = o^A \iota^{A'},\ \bar{m}^a = \iota^A o^{A'}$. The metric can be expressed in terms of the spin-frame via the epsilon spinor, ε_{AB}, or in terms of the tetrad:

$$g_{ab} = \varepsilon_{AB}\varepsilon_{A'B'} = l_a n_b + n_a l_b - m_a \bar{m}_b - \bar{m}_a m_b \,,$$

The components of the metric in this tetrad take the form

$$g_{\boldsymbol{ab}} = n_{\boldsymbol{ab}} \coloneqq \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix} . \tag{3.1}$$

Once again utilising the unique covariant derivative operator (Levi-Civita) that is both torsion free, and metric preserving, let us denote the four directional tetrad covariant derivative operators as $(l^a\nabla_a,\, n^a\nabla_a,\, m^a\nabla_a,\, \bar{m}^a\nabla_a) = (D,\, \Delta,\, \delta,\, \bar{\delta})$. We would also like it to work with complex objects, and define $\nabla_a(f + ig) = \nabla_a(f) + i\nabla_a(g)$ for real functions f, g. This means that $\overline{\nabla_a h} = \nabla_a \bar{h}$. When required, this can be extended to operate on spinors, defined as

[1] This means that each of the tetrad vectors are null with respect to the metric, $l^a l_a = n^a n_a = m^a m_a = \bar{m}^a \bar{m}_a = 0$.

a map $\nabla_{AA'} = \nabla_a : \mathcal{G}^B \to \mathcal{G}^B_{AA'}$ satisfying linearity over the scalars, which then extends to higher valence spinors in the usual way as for tensors, assuming a Leibniz law shall hold for contracted products. Using the Levi-Civita covariant derivative we will also have the property $\nabla_a \varepsilon_{AB} = 0$. We will adopt the sign convention so that the Ricci identity reads

$$2\nabla_{[a}\nabla_{b]}Z^c = R^c{}_{dab}Z^d\,,$$

and the Weyl tensor, that characterizes curvature in the absence of matter, is given by the trace free part of the Riemann tensor:

$$C_{ab}{}^{cd} = R_{ab}{}^{cd} - 2R_{[a}{}^{[c}g_{b]}{}^{d]} + \frac{1}{3}Rg_{[a}{}^{c}g_{b]}{}^{d}\,,$$

where R_{ab}, R are the Ricci tensor and Ricci scalar respectively [6]. We will find particularly useful, the decomposition of the Riemann tensor into totally symmetric spinor parts. We find (see Appendix A.5)

$$\begin{aligned} R_{abcd} = R_{AA'BB'CC'DD'} = \Psi_{ABCD}\varepsilon_{A'B'}\varepsilon_{C'D'} + \overline{\Psi}_{A'B'C'D'}\varepsilon_{AB}\varepsilon_{CD} \\ +\Phi_{ABC'D'}\varepsilon_{A'B'}\varepsilon_{CD} + \overline{\Phi}_{A'B'CD}\varepsilon_{AB}\varepsilon_{C'D'} \\ +2\Lambda\left(\varepsilon_{AC}\varepsilon_{BD}\varepsilon_{A'C'}\varepsilon_{B'D'} - \varepsilon_{AD}\varepsilon_{BC}\varepsilon_{A'D'}\varepsilon_{B'C'}\right) \end{aligned}$$

where $\Psi_{ABCD} = \Psi_{(ABCD)}$ is the Weyl spinor, $\Phi_{ABC'D'} = \Phi_{(AB)(C'D')}$ is the Ricci spinor, and $\Lambda = R/24$ gives the scalar curvature. There are five independent dyad components of the Weyl tensor, and we will introduce notation to denote them as

$$\begin{aligned} &\Psi_0 = \Psi_{0000}\,, \quad \Psi_1 = \Psi_{0001}\,, \quad \Psi_2 = \Psi_{0011}\,, \\ &\Psi_3 = \Psi_{0111}\,, \quad \Psi_4 = \Psi_{1111}\,. \end{aligned}$$

We also introduce analogous notation[2] for the nine independent dyad components of the Ricci spinor;

$$\begin{aligned} &\Phi_{00} = \Phi_{000'0'}\,, \quad \Phi_{10} = \Phi_{010'0'}\,, \quad \Phi_{20} = \Phi_{110'0'}\,, \\ &\Phi_{01} = \Phi_{000'1'}\,, \quad \Phi_{11} = \Phi_{010'1'}\,, \quad \Phi_{21} = \Phi_{110'1'}\,, \\ &\Phi_{02} = \Phi_{001'1'}\,, \quad \Phi_{12} = \Phi_{011'1'}\,, \quad \Phi_{22} = \Phi_{111'1'}\,. \end{aligned}$$

These components are also representable in terms of the tetrad components of the Weyl tensor and Ricci tensor, via the decomposition of the Weyl tensor

$$-C_{abcd} = \Psi_{ABCD}\varepsilon_{A'B'}\varepsilon_{C'D'} + \overline{\Psi}_{A'B'C'D'}\varepsilon_{AB}\varepsilon_{CD}\,,$$

and the relationship between the Ricci tensor and Ricci spinor

$$R_{ab} - \frac{1}{4}Rg_{ab} = -2\Phi_{ab} = -2\Phi_{ABA'B'}\,.$$

[2]Note these are not the tetrad components of $\Phi_{ab} = \Phi_{AA'BB'}$, for example $\Phi_{01} = \Phi_{ABC'D'}o^A o^B o^{C'}\iota^{D'} = \Phi_{ABA'B'}o^A o^B o^{A'}\iota^{B'} = \Phi_{ab}l^a m^b \neq \Phi_{ab}l^a n^b$.

3.1 Spin-coefficients and spin-coefficient equations

The spin-coefficients are a set of complex scalars that constitute the base variables in the spin-coefficient formalism. Denoting the tetrad basis with $g^a_{\boldsymbol{a}}$, the components of the covariant derivatives of the tetrad covectors are known as the Ricci rotation coefficients, denoted via $\gamma^{\boldsymbol{a}}{}_{\boldsymbol{bc}} = g^a_{\boldsymbol{b}} \nabla_{\boldsymbol{c}} g^{\boldsymbol{a}}_a = g^a_{\boldsymbol{b}} g^b_{\boldsymbol{c}} \nabla_b g^{\boldsymbol{a}}_a$. The rotation coefficients have a symmetry $\gamma_{\boldsymbol{abc}} = -\gamma_{\boldsymbol{bac}} = n_{\boldsymbol{ad}} \gamma^{\boldsymbol{d}}{}_{\boldsymbol{bc}}$, so that there are only 12 independent components, if taken to be complex. Denoting these in Table 3.1, they constitute the spin coefficients. They are so named for their alternate

$$
\begin{aligned}
&\pi = -\gamma_{130}\,, && \lambda = -\gamma_{133}\,, && \mu = -\gamma_{132}\,, && \nu = -\gamma_{131}\,,\\
&\epsilon = \frac{1}{2}(\gamma_{010} - \gamma_{230})\,, && \alpha = \frac{1}{2}(\gamma_{013} - \gamma_{233})\,, && \beta = \frac{1}{2}(\gamma_{012} - \gamma_{232})\,, && \gamma = \frac{1}{2}(\gamma_{011} - \gamma_{231})\,,\\
&\kappa = \gamma_{020}\,, && \rho = \gamma_{023}\,, && \sigma = \gamma_{022}\,, && \tau = \gamma_{021}\,.
\end{aligned}
$$

Table 3.1: The 12 spin coefficients as given in terms of the tetrad.

definition in terms of the spin-frame; taking now the components of the covariant derivatives of the spin-frame dyad dual basis, denoted as $\gamma_{\boldsymbol{AA'C}}{}^{\boldsymbol{B}} = \varepsilon_A{}^{\boldsymbol{B}} \nabla_{\boldsymbol{AA'}} \varepsilon_{\boldsymbol{C}}{}^A$, where $\varepsilon_{\boldsymbol{A}}{}^A$, $\varepsilon_A{}^{\boldsymbol{A}}$ denote the basis and dual basis on the spin frame. When lowered, it has symmetry $\gamma_{\boldsymbol{AA'CB}} = \gamma_{\boldsymbol{AA'BC}}$, and we assign these components as the spin coefficients in the manner given in Table 3.2. To see how these definitions are equivalent we can use the definitions of the tetrad vectors in terms of the spin-frame and the Leibniz rule for the covariant derivative. At this stage let

$$
\begin{aligned}
&\pi = \gamma_{00'11}\,, && \lambda = \gamma_{10'11}\,, && \mu = -\gamma_{01'11}\,, && \nu = \gamma_{11'11}\,,\\
&\epsilon = \gamma_{00'10}\,, && \alpha = \gamma_{10'10}\,, && \beta = \gamma_{01'10}\,, && \gamma = \gamma_{11'10}\,,\\
&\kappa = \gamma_{00'00}\,, && \rho = \gamma_{10'00}\,, && \sigma = \gamma_{01'00}\,, && \tau = \gamma_{11'00}\,.
\end{aligned}
$$

Table 3.2: The 12 spin coefficients as given in terms of the spin basis.

us introduce an alternate notation in terms of a *prime* operation. If we let the priming of a symbol denote the quantity under a transformation

$$
o^A \to i\iota^A\,, \quad \iota^A \to io^A\,. \tag{3.2}
$$

then we find an equivalence between some of the spin coefficients under the prime operation. For example $\pi = -\tau'$. Priming commutes with complex conjugation, addition and multiplication, and we also note that

$$
(l^a)' = n^a\,, \quad (n^a)' = l^a\,, \quad (m^a)' = \bar{m}^a\,, \quad (\bar{m}^a)' = m^a\,.
$$

In general, priming quantities related to l^a gives an equivalent quantity for n^a, while priming quantities related to m^a gives ones that are equivalent to quantities relating to $\bar{m}^a$, being essentially the same as conjugation in these cases, we note that $D' = \Delta$, while $\delta' = \bar{\delta}$. The *spin coefficient equations* (or *Newman-Penrose equations*) are then a set of equations in terms of the spin coefficients and quantities related to the Riemann tensor that constitute the field equations for the spin coefficients. Written explicitly, we find as given in Newman

and Penrose [7]:

$$
\begin{aligned}
&D\rho - \bar{\delta}\kappa = \rho^2 + \sigma\bar{\sigma} + (\epsilon + \bar{\epsilon})\,\rho - \bar{\kappa}\tau - \kappa\left(3\alpha + \bar{\beta} - \pi\right) + \Phi_{00}\\
&D\sigma - \delta\kappa = (\rho + \bar{\rho})\,\sigma + (3\epsilon - \bar{\epsilon})\,\sigma - (\tau - \bar{\pi} + \bar{\alpha} + 3\beta)\,\kappa + \Psi_0\\
&D\tau - \Delta\kappa = (\tau + \bar{\pi})\,\rho + (\bar{\tau} + \pi)\,\sigma + (\epsilon - \bar{\epsilon})\,\tau - (3\gamma + \bar{\gamma})\,\kappa + \Psi_1 + \Phi_{01}\\
&D\alpha - \bar{\delta}\epsilon = (\rho + \bar{\epsilon} - 2\epsilon)\,\alpha + \beta\bar{\sigma} - \bar{\beta}\epsilon - \kappa\lambda - \bar{\kappa}\gamma + (\epsilon + \rho)\,\pi + \Phi_{10}\\
&D\beta - \delta\epsilon = (\alpha + \pi)\,\sigma + (\bar{\rho} - \bar{\epsilon})\,\beta - (\mu + \gamma)\,\kappa - (\bar{\alpha} - \bar{\pi})\,\epsilon + \Psi_1\\
&D\gamma - \Delta\epsilon = (\tau + \bar{\pi})\,\alpha + (\bar{\tau} + \pi)\,\beta - (\epsilon + \bar{\epsilon})\,\gamma - (\gamma + \bar{\gamma})\,\epsilon + \tau\pi - \nu\kappa + \Psi_2 - \Lambda + \Psi_{11}\\
&D\lambda - \bar{\delta}\pi = \rho\lambda + \bar{\sigma}\mu + \pi^2 + \left(\alpha - \bar{\beta}\right)\pi - \nu\bar{\kappa} - (3\epsilon - \bar{\epsilon})\,\lambda + \Psi_{20}\\
&D\mu - \delta\pi = \bar{\rho}\mu\sigma\lambda + \pi\bar{\pi} - (\epsilon + \bar{\epsilon})\,\mu - (\bar{\alpha} - \beta)\,\pi - \nu\kappa + \Psi_2 + 2\Lambda\\
&D\nu - \Delta\pi = (\pi + \bar{\tau})\,\mu + (\bar{\pi} + \tau)\,\lambda + (\gamma - \bar{\gamma})\,\pi - (3\epsilon + \bar{\epsilon})\,\nu + \Psi_3 + \Phi_{21}\\
&\Delta\lambda - \bar{\delta}\nu = -(\mu + \bar{\mu})\,\lambda - (3\gamma - \bar{\gamma})\,\lambda + \left(3\alpha + \bar{\beta} + \pi - \bar{\tau}\right)\nu - \Psi_4\\
&\delta\rho - \bar{\delta}\sigma = (\bar{\alpha} + \beta)\,\rho - \left(3\alpha - \bar{\beta}\right)\sigma + (\rho - \bar{\rho})\,\tau + (\mu - \bar{\mu})\,\kappa - \Psi_1 + \Phi_{01}\\
&\delta\alpha - \bar{\delta}\beta = \mu\rho - \lambda\sigma + \alpha\bar{\alpha} + \beta\bar{\beta} - 2\alpha\beta + (\rho - \bar{\rho})\,\gamma + (\mu - \bar{\mu})\,\epsilon - \Psi_2 + \Lambda + \Phi_{11}\\
&\delta\lambda - \bar{\delta}\mu = (\rho - \bar{\rho})\,\nu + (\mu - \bar{\mu})\,\pi + \left(\alpha + \bar{\beta}\right)\mu + (\bar{\alpha} - 3\beta)\,\lambda - \Psi_3 + \Phi_{21}\\
&\delta\nu - \Delta\mu = \mu^2 + \lambda\bar{\lambda} + (\gamma + \bar{\gamma})\,\mu - \nu\pi + (\tau - 3\beta - \bar{\alpha})\,\nu + \Phi_{22}\\
&\delta\gamma - \Delta\beta = (\tau - \bar{\alpha} - \beta)\,\gamma + \mu\tau - \sigma\nu - \epsilon\bar{\nu} - (\gamma - \bar{\gamma} - \mu)\,\beta + \alpha\bar{\lambda} + \Phi_{12}\\
&\delta\tau - \Delta\sigma = \mu\sigma + \bar{\lambda}\rho + (\tau + \beta - \bar{\alpha})\,\tau - (3\gamma - \bar{\gamma})\,\sigma - \kappa\bar{\nu} + \Phi_{02}\\
&\Delta\rho - \bar{\delta}\tau = -\rho\bar{\mu} - \sigma\lambda + \left(\bar{\beta} - \alpha - \bar{\tau}\right)\tau + (\gamma + \bar{\gamma})\,\rho + \nu\kappa - \Psi_2 - 2\Lambda\\
&\Delta\alpha - \bar{\delta}\gamma = (\rho + \epsilon)\,\nu - (\tau + \beta)\,\lambda + (\bar{\gamma} - \bar{\mu})\,\alpha + \left(\bar{\beta} - \bar{\tau}\right)\gamma - \Psi_3\,.
\end{aligned}
\tag{3.3}
$$

These equations are simply derived from the commutator relations for the covariant derivative acting on a spin-vector. The commutation relations for the covariant derivative acting on scalars take the form

$$
\begin{aligned}
&\Delta D - D\Delta = (\gamma + \bar{\gamma})\,D + (\epsilon + \bar{\epsilon})\,\Delta - (\tau + \bar{\pi})\,\bar{\delta} - (\bar{\tau} + \pi)\,\delta\,,\\
&\delta D - D\delta = (\bar{\alpha} + \beta - \bar{\pi})\,D + \kappa\Delta - \sigma\bar{\delta} - (\bar{\rho} + \epsilon - \bar{\epsilon})\,\delta\,,\\
&\delta\Delta - \Delta\delta = -\bar{\nu}D + (\tau - \bar{\alpha} - \beta)\,\Delta + \bar{\lambda}\bar{\delta} + (\mu - \gamma + \bar{\gamma})\,\delta\,,\\
&\bar{\delta}\delta - \delta\bar{\delta} = (\bar{\mu} - \mu)\,D + (\bar{\rho} - \rho)\,\Delta - (\bar{\alpha} - \beta)\,\bar{\delta} - \left(\bar{\delta} - \alpha\right)\delta\,.
\end{aligned}
\tag{3.4}
$$

We will also find useful the equations for the *transport equations* that relate the covariant derivatives of each of the tetrad vectors to the spin coefficients

$$\begin{aligned}
Dl^a &= (\epsilon + \bar{\epsilon})\, l^a - \bar{\kappa} m^a - \kappa \bar{m}^a \,, \\
\Delta l^a &= (\gamma + \bar{\gamma})\, l^a - \bar{\tau} m^a - \tau \bar{m}^a \,, \\
\delta l^a &= (\bar{\alpha} + \beta)\, l^a - \bar{\rho} m^a - \sigma \bar{m}^a \,, \\
\bar{\delta} l^a &= \left(\alpha + \bar{\beta}\right) l^a - \bar{\sigma} m^a - \rho \bar{m}^a \,, \\
Dn^a &= -\left(\epsilon + \bar{\epsilon}\right) n^a + \pi m^a + \bar{\pi} \bar{m}^a \,, \\
\Delta n^a &= -\left(\gamma + \bar{\gamma}\right) n^a + \nu m^a + \bar{\nu} \bar{m}^a \,, \\
\delta n^a &= -\left(\bar{\alpha} + \beta\right) n^a + \mu m^a + \bar{\lambda} \bar{m}^a \,, \\
\bar{\delta} n^a &= -\left(\alpha + \bar{\beta}\right) n^a + \lambda m^a + \bar{\mu} \bar{m}^a \,, \\
Dm^a &= (\epsilon - \bar{\epsilon})\, m^a + \bar{\pi} l^a - \kappa n^a \,, \\
\Delta m^a &= (\gamma - \bar{\gamma}) + \bar{\nu} l^a - \tau n^a \,, \\
\delta m^a &= (\beta - \bar{\alpha})\, m^a + \bar{\lambda} l^a - \sigma n^a \,, \\
\bar{\delta} m^a &= \left(\alpha - \bar{\beta}\right) m^a + \bar{\mu} l^a - \rho n^a \,, \\
D\bar{m}^a &= (\bar{\epsilon} - \epsilon)\, \bar{m}^a + \pi l^a - \bar{\kappa} n^a \,, \\
\Delta \bar{m}^a &= (\bar{\gamma} - \gamma)\, \bar{m}^a + \nu l^a - \bar{\tau} n^a \,, \\
\delta \bar{m}^a &= (\bar{\alpha} - \beta)\, \bar{m}^a + \mu l^a - \bar{\rho} n^a \,, \\
\bar{\delta} \bar{m}^a &= \left(\bar{\beta} - \alpha\right) \bar{m}^a + \lambda l^a - \bar{\sigma} n^a
\end{aligned} \tag{3.5}$$

given from O'Donnell [8]. Finally we note that the Bianchi identities may be translated into the notation of the spin coefficients, but the full equations are rather unwieldy. For the full details see pages 62-64 of O'Donnell [8].

3.1.1 Weighted quantities

Another feature of the spin-coefficient formalism is introduced in Penrose and Rindler [5], and concerns *weighted quantities*; where $(o^{\mathcal{A}}, \iota^{\mathcal{A}})$ is a spin-frame we consider change of bases where the directions and inner product are left invariant. These have the general form

$$o^{\mathcal{A}} \to \lambda o^{\mathcal{A}} \,, \quad \iota^{\mathcal{A}} \to \lambda^{-1} \iota^{\mathcal{A}} \,, \tag{3.6}$$

or in terms of the induced null tetrad

$$l^a \to \lambda \bar{\lambda} l^a \,, \quad n^a \to \lambda^{-1} \bar{\lambda}^{-1} n^a \,, \quad m^a \to \lambda \bar{\lambda}^{-1} m^a \,, \quad \bar{m}^a \to \bar{\lambda} \lambda^{-1} \bar{m}^a \,.$$

If we define $\lambda^2 = Re^{i\theta}$, for R, θ real, we have

$$l^a \to R l^a \,, \quad n^a \to R^{-1} n^a \,, \quad m^a \to e^{i\theta} m^a \,, \quad \bar{m}^a \to e^{i\theta} \bar{m}^a \,,$$

which describes a boost with a rotation. These types of transformations form a group and have a rich development. Let us call scalars, η, that transform under a (3.6) with the form

$$\eta \to \lambda^p \bar{\lambda}^q \eta \,, \tag{3.7}$$

weighted quantities of type $\{p, q\}$. We can alternately describe them in terms of *spin-weight* and *boost-weight*, given by $\frac{1}{2}(p-q)$ and $\frac{1}{2}(p+q)$ respectively. If we recast (3.7) in terms of $Re^{i\theta}$ we have

$$\eta \to R^{\frac{1}{2}(p+q)} e^{i\theta \frac{1}{2}(p-q)} \eta = R^w e^{is\theta} \eta \,,$$

for spin weight s, and boost weight w, with the definitions now appearing more natural. The spin coefficients themselves can be divided into those that are weighted, and those that are not. It is possible to work with only weighted quantities in a self contained algebra, where the product of two weighted quantities produce a weighted quantity of the type of the sums of each p and q, and only those weighted quantities of the same type may be added. Complex conjugation takes a quantity of type $\{p, q\}$ to one of type $\{q, p\}$, while the priming operation of (3.2) changes $\{p, q\}$ to $\{-p, -q\}$. To allow for derivatives in the contained algebra we change the scalar derivatives to mean that derivatives on weighted quantities also produce weighted quantities. Let us take

$$\begin{aligned} \text{þ}\eta &= (D - p\epsilon - q\bar{\epsilon})\,\eta \,, \\ \text{ð}\eta &= (\delta - p\beta - q\bar{\alpha})\,\eta \,, \end{aligned} \tag{3.8}$$

as *weighted derivatives*, then we have that

$$\text{ð}' = \bar{\text{ð}} \,, \quad \bar{\text{þ}} = \text{þ} \,,$$

and these operators satisfy the usual properties of differential operators along with the Leibniz rule. These are chosen specifically so as to cancel those terms introduced by the derivative operators that are problematic. Now the operators (ð, þ) have types ($\{1, -1\}$, $\{1, 1\}$) respectively in the sense that they change the type of a weighted quantity by the amounts listed. It is possible to recast the entire formulation of spin coefficients to deal with only weighted quantities, and to deal with those that are not naturally weighted, we would introduce the weighted derivative operators into the spin coefficient equations and these terms are eliminated. We will not expand on this weighted quantity formalism in this report, but will use the weighted derivatives as shorthand notation.

3.2 Coordinates, conditions and assumptions

Let us now set up a coordinate system as in Section 2, $(u, \tilde{r}, x^{\mathcal{A}})$, where instead of r, the non-affine parameter along hypersurfaces, we use an affine parameter $\tilde{r}$. The parameter set up in Section 2 is sometimes referred to as a *luminosity distance* as used in Bondi, Van der Burg, and Metzner [9], it is related to an affine parameter $\tilde{r}$ via $\partial_r \tilde{r} = e^{2\beta}$. We then would like to choose our tetrad vectors so as to be fitting for the coordinates. Let l^a be tangent to the null geodesics labelled by u and (affinely) parametrized by $\tilde{r}$. In the language of spin-coefficients, we can use (3.5) to see in order that $Dl^a = \alpha l^a$ we require $\kappa = 0$, and $\epsilon + \bar{\epsilon} = 0$ gives affine normalization. The vector l^a is now essentially[3] the same as that introduced in Section 2. We also note that the vectors $m^a, \bar{m}^a$ are related to the vectors introduced in Section 2.1 by a factor of r, as $M^a, \bar{M}^a$ are normalised with respect to the 2-metric $h^{\mathcal{AB}} = r^2 g^{\mathcal{AB}}$, but the

[3]Remember that here we use abstract index notation, whereas in Section 2 we do not, but they represent the same vector.

change in coordinates to an affine $\tilde{r}$ also affects the other components. Then we take the other tetrad vectors to be parallelly propagated along the geodesics, or in other words set each of the $Dg^a_{\boldsymbol{a}} = 0$. This has a simple meaning in the spin coefficients of $\epsilon - \bar{\epsilon} = 0 = \pi$, seen from (3.5). Collecting these results, the spin coefficients simplify; $\epsilon = \pi = \kappa = 0$. Finally, we note that as the foliation is a gradient field (via Section 2, $l_{\boldsymbol{a}} = \partial_{\boldsymbol{a}} u$) we have $\rho = \bar{\rho}$, and $\tau = \bar{\alpha} + \beta$. In these coordinates the metric has form

$$g^{\boldsymbol{ab}} = \begin{bmatrix} 0 & 1 & 0^{\mathcal{A}} \\ 1 & 2(U - \omega\bar{\omega}) & X^{\mathcal{A}} - \left(\bar{\omega}\zeta^{\mathcal{A}} + \bar{\omega}\zeta^{\mathcal{A}}\right) \\ 0^{\mathcal{B}} & X^{\mathcal{A}} - \left(\bar{\omega}\zeta^{\mathcal{A}} + \bar{\omega}\zeta^{\mathcal{A}}\right) & -\left(\zeta^{\mathcal{A}}\bar{\zeta}^{\mathcal{B}} + \bar{\zeta}^{\mathcal{A}}\zeta^{\mathcal{B}}\right) \end{bmatrix},$$

where $U, \omega, X^{\mathcal{A}}, \zeta^{\mathcal{A}}$ are arbitrary functions of the coordinates such that in coordinate components

$$l^{\boldsymbol{a}} = (0, 1, 0^{\mathcal{A}}), \quad n^{\boldsymbol{a}} = (1, U, X^{\mathcal{A}}), \quad m^{\boldsymbol{a}} = (0, \omega, \zeta^{\mathcal{A}}).$$

In order to obey the tetrad relationships, the tetrad vectors must have this form. The directional derivatives will have the form (acting on scalars)

$$\begin{aligned} D &= l^{\boldsymbol{a}}\partial_{\boldsymbol{a}} = \frac{\partial}{\partial\tilde{r}}, \\ \Delta &= n^{\boldsymbol{a}}\partial_{\boldsymbol{a}} = \frac{\partial}{\partial u} + U\frac{\partial}{\partial\tilde{r}} + X^{\mathcal{A}}\frac{\partial}{\partial x^{\mathcal{A}}}, \\ \delta &= m^{\boldsymbol{a}}\partial_{\boldsymbol{a}} = \omega\frac{\partial}{\partial\tilde{r}} + \zeta^{\mathcal{A}}\frac{\partial}{\partial x^{\mathcal{A}}}, \\ \bar{\delta} &= \bar{m}^{\boldsymbol{a}}\partial_{\boldsymbol{a}} = \bar{\omega}\frac{\partial}{\partial\tilde{r}} + \bar{\zeta}^{\mathcal{A}}\frac{\partial}{\partial x^{\mathcal{A}}}. \end{aligned}$$

Going forward we will also assume that the cosmological constant is zero and the vacuum space equations hold; In terms of the spinor decomposition of the Riemann tensor this means

that $\Phi_{ab} = 0 = \Lambda$. The spin coefficient field equations are then simplified drastically:

$$
\begin{aligned}
D\rho &= \rho^2 + \sigma\bar{\sigma}\,, \\
D\sigma &= 2\rho\sigma + \Psi_0\,, \\
D\tau &= \tau\rho + \bar{\tau}\sigma + \Psi_1\,, \\
D\alpha &= \rho\alpha + \beta\bar{\sigma}\,, \\
D\beta &= \alpha\sigma + \rho\beta + \Psi_1\,, \\
D\gamma &= \tau\alpha + \bar{\tau}\beta + \Psi_2\,, \\
D\lambda &= \rho\lambda + \bar{\sigma}\mu\,, \\
D\mu &= \rho\mu + \sigma\lambda + \Psi_2\,, \\
D\nu &= \bar{\tau}\mu + \tau\lambda + \Psi_3\,, \\
\Delta\lambda - \bar{\delta}\nu &= (\bar{\gamma} - \mu - \bar{\mu} - 3\gamma)\,\lambda + 2\alpha\nu - \Psi_4\,, \\
\delta\rho - \bar{\delta}\sigma &= \rho\tau - \left(3\alpha - \bar{\beta}\right)\sigma - \Psi_1\,, \\
\delta\alpha - \bar{\delta}\beta &= \mu\rho - \lambda\sigma + \alpha\bar{\alpha} + \beta\bar{\beta} - 2\alpha\beta - \Psi_2\,, \\
\delta\lambda - \bar{\delta}\mu &= \mu\bar{\tau} + \lambda\left(\alpha - 3\beta\right) - \Psi_3\,, \\
\delta\nu - \Delta\mu &= \mu^2 + \lambda\bar{\lambda} + \mu\left(\gamma + \bar{\gamma}\right) - 2\beta\nu\,, \\
\delta\gamma - \Delta\beta &= \mu\tau - \sigma\nu - \epsilon\bar{\nu} - (\gamma - \bar{\gamma} - \mu)\,\beta + \alpha\bar{\lambda}\,, \\
\delta\tau - \Delta\sigma &= \mu\sigma + \rho\bar{\lambda} + 2\tau\beta - (3\gamma - \bar{\gamma})\,, \\
\Delta\rho - \bar{\delta}\tau &= -2\tau\alpha + (\gamma + \bar{\gamma} - \bar{\mu})\,\rho - \Psi_2\,, \\
\Delta\alpha - \bar{\delta}\gamma &= \nu\rho - (\tau + \beta)\,\lambda + (\bar{\gamma} - \gamma - \bar{\mu})\,\alpha - \Psi_3\,,
\end{aligned}
\tag{3.9}
$$

and we may write the Bianchi identities more simply as

$$
\begin{aligned}
\bar{\delta}\Psi_0 - D\Psi_1 &= 4\alpha\Psi_0 - 4\rho\Psi_1\,, \\
\bar{\delta}\Psi_0 - D\Psi_2 &= \lambda\Psi_0 + 2\alpha\Psi_1 - 3\rho\Psi_2\,, \\
\bar{\delta}\Psi_2 - D\Psi_3 &= 2\lambda\Psi_1 - 2\rho\Psi_3\,, \\
\bar{\delta}\Psi_3 - D\Psi_4 &= 3\lambda\Psi_2 - 2\alpha\Psi_3 - \rho\Psi_4\,, \\
\Delta\Psi_0 - \delta\Psi_1 &= (4\gamma - \mu)\,\Psi_0 - (4\tau + 2\beta)\,\Psi_1 + 3\sigma\Psi_2\,, \\
\Delta\Psi_1 - \delta\Psi_2 &= \nu\Psi_0 + 2\,(\gamma - \mu)\,\Psi_1 - 3\tau\Psi_2 + 2\sigma\Psi_3\,, \\
\Delta\Psi_2 - \delta\Psi_3 &= 2\nu\Psi_1 - 3\mu\Psi_2 + 2\,(\beta - \tau)\,\Psi_3 + \sigma\Psi_4\,, \\
\Delta\Psi_3 - \delta\Psi_4 &= 3\nu\Psi_2 - (2\gamma + 4\mu)\,\Psi_3 + (4\beta - \tau)\,\Psi_4\,.
\end{aligned}
\tag{3.10}
$$

Finally, applying the commutation relations, (3.4), to each of the coordinates $u, \tilde{r}, x^{\mathcal{A}}$ we can find the relationship between the metric functions and the spin coefficients:

$$
\begin{aligned}
DU &= \tau\bar{\omega} + \bar{\tau}\omega - (\gamma + \bar{\gamma})\,, \\
DX^{\mathcal{A}} &= \tau\bar{\zeta}^{\mathcal{A}} + \bar{\tau}\zeta^{\mathcal{A}}\,, \\
D\omega &= \sigma\bar{\omega} + \rho\omega - \tau\,, \\
D\zeta^{\mathcal{A}} &= \sigma\bar{\zeta}^{\mathcal{A}} + \rho\zeta^{\mathcal{A}}\,, \\
\delta U - \Delta\omega &= -\bar{\nu} + \bar{\lambda}\bar{\omega} + (\mu - \gamma + \bar{\gamma})\,\omega\,, \\
\delta X^{\mathcal{A}} - \Delta\zeta^{\mathcal{A}} &= \bar{\lambda}\zeta^{\mathcal{A}} + (\mu - \gamma + \bar{\gamma})\,\zeta^{\mathcal{A}}\,, \\
\bar{\delta}\omega - \delta\bar{\omega} &= \bar{\mu} - \mu - (\bar{\alpha} - \beta)\,\bar{\omega} + \left(\alpha - \bar{\beta}\right)\omega\,, \\
\bar{\delta}\zeta^{\mathcal{A}} - \delta\bar{\zeta}^{\mathcal{A}} &= -\,(\bar{\alpha} - \beta)\,\bar{\zeta}^{\mathcal{A}} + \left(\alpha - \bar{\beta}\right)\bar{\zeta}^{\mathcal{A}}\,.
\end{aligned}
\tag{3.11}
$$

With these conditions in place, we can put in the asymptotic flatness as $\tilde{r} \to \infty$. Whenever necessary, we will denote the components of a $1/\tilde{r}$ expansion that are independent of $\tilde{r}$ with numbered superscript. We will start by assuming that the Weyl tensor has fall off $\Psi_0 = \Psi_0^0 \tilde{r}^{-5} + O(\tilde{r}^{-6})$.

3.2.1 Geometric interpretations

One advantage of the Newman-Penrose formalism is that some of the spin coefficients have simple geometric interpretations. Using the equations (3.5), we have already seen that κ is related to the first curvature for the congruence with tangent vector l^a, as $\kappa = 0$ makes this tangent to a geodesic, while $\mathrm{Re}(\epsilon)$ dictates the geodesic proportionality. When it is the case that $Dl^a = 0$, the spin coefficient ρ can be given by

$$\rho = \frac{1}{2}\left(-\nabla_a l^a + i\sqrt{\nabla_{[a} l_{b]} \nabla^a l^b}\right) = -\frac{1}{2}\nabla_a l^a \,,$$

while σ satisfies

$$\sigma\bar{\sigma} = \frac{1}{2}\left(\nabla_{(a} l_{b)} \nabla^a l^b - \frac{1}{2}\left(\nabla_a l^a\right)^2\right) = \frac{1}{2}\nabla_a l_b \nabla^a l^b - \rho^2 \,,$$

from the definitions [10], where the second equality in each is given only in the case where the foliation is a gradient field. They represent the (complex) divergence and shear of the null ray congruence given by l^a [10]. Finally, τ describes the change in direction of l^a as we go toward n^a, again through (3.5). We also find that $\left(-\bar{\nu}, -Re(\gamma), -\bar{\mu}, -\bar{\lambda}, -\bar{\pi}\right)$ are respectively the equivalent spin coefficients to $(\kappa, Re(\epsilon), \rho, \sigma, \tau)$ for congruences given by n^a, with the same geometric interpretations.

3.3 Solution and uniqueness for asymptotically flat case

With the relevant assumptions given in Section 3.2, we may solve for the asymptotic dependence of each of the spin coefficients, Weyl tensor components and relevant metric variables, given suitable initial data. The field equations (3.9) can be solved with a systematic procedure, beginning with the radial equations relating to the $D = l^a \nabla_a$ derivative. The first two equations,

$$D\rho = \rho^2 + \sigma\bar{\sigma} \,, \quad D\sigma = 2\rho\sigma + \Psi_0 \,,$$

can be transformed into a linearised matrix system and solved with standard methods to find the asymptotic dependence of ρ, the divergence [7]. From here the integration proceeds through the equations to find the asymptotic dependence of all the relevant variables, beginning with the radial equations field and metric equations involving D derivatives (3.9), (3.11). These allow us to solve for the asymptotic behaviour spin coefficients and metric variables, along with the asymptotic dependence on each of the Weyl tensor components that follows a distinct pattern;

$$\Psi_n = \Psi_n^0 \tilde{r}^{-5+n} + O(\tilde{r}^{-6+n}), \quad (n = 1, 2, 3, 4) \,.$$

This result for the Weyl tensor components is known as the *peeling theorem*, as the components of the Weyl tensor 'peel off', that is to say decay into negligibility in order, as we go along the null geodesics parametrized by the $\tilde{r}$ coordinate. For a unique solution to the field equations we require some free functions to be given as inputs, we take the following as the most natural initial data:

- The first coefficient in a $1/\tilde{r}$ expansion of the Weyl tensor component Ψ_0 on an initial hypersurface defined by $u = u_0$, denoted by $\Psi_0^0(u_0, \zeta, \bar{\zeta})$.
- The first coefficient of the Weyl tensor component Ψ_1 on $u = u_0$, denoted by $\Psi_1^0(u_0, \zeta, \bar{\zeta})$.
- The first coefficient of the Weyl tensor component Ψ_2 on $u = u_0$, denoted by $\Psi_2^0(u_0, \zeta, \bar{\zeta})$.
- The function $\sigma^0(u, \zeta, \bar{\zeta})$ known as the *Bondi shear*, which is the first coefficient in the $1/\tilde{r}$ expansion of σ.

We are then able to give the relevant quantities in terms of these values. By using the remaining non-radial field and metric equations (3.9), (3.11), we find on inspection through the powers of $\tilde{r}$ the integration functions. In particular we will use

$$\Psi_3^0 = \eth \dot{\bar{\sigma}}^0 \,, \tag{3.12}$$

$$\Psi_4^0 = -\ddot{\bar{\sigma}}^0 \,, \tag{3.13}$$

where the weighted derivative $\eth$ is as defined in (3.8) and a dot denotes a u derivative. Also found is the equation

$$\bar{\eth}^2 \sigma - \eth^2 \bar{\sigma} = \Psi_2^0 - \bar{\Psi}_2^0 + \sigma^0 \dot{\bar{\sigma}}^0 - \bar{\sigma}^0 \dot{\sigma}^0 \,,$$

which, if we define $\mathcal{M} \coloneqq \Psi_2^0 + \bar{\sigma}^0 \dot{\sigma}^0 + \eth^2 \bar{\sigma}^0$, simplifies to $\bar{\mathcal{M}} = \mathcal{M}$. Known as the mass aspect in some texts [11], we will refer to this quantity as the *modified mass aspect* to distinguish it from the mass aspect as defined in Section 2. The Bianchi identities (3.10) give the evolution over time of the initial data functions, and thus by proxy, of the entire system. We find, in particular, that

$$\dot{\Psi}_2^0 = -\eth \Psi_3^0 + \sigma^0 \Psi_4^0 = -\eth^2 \dot{\bar{\sigma}}^0 - \sigma^0 \ddot{\bar{\sigma}}^0 = -\partial_u (\sigma^0 \dot{\bar{\sigma}}^0) - \eth^2 \dot{\bar{\sigma}}^0 + \dot{\sigma}^0 \dot{\bar{\sigma}}^0 \,,$$

via (3.12) and (3.13), which gives

$$\dot{\mathcal{M}} = \dot{\sigma}^0 \dot{\bar{\sigma}}^0 \,. \tag{3.14}$$

Using (3.14), it can be shown that the Bondi mass loss theorem holds.

$$\dot{m}(u) = -\oint \dot{\sigma}^0 \dot{\bar{\sigma}}^0 \, dS \,,$$

the integral taken over the 2-sphere.

Chapter 4

Connections Between Definitions and Interpretations

The Bondi-Sachs and Newman-Penrose formalisms use different language to describe the Einstein equations and their solutions. In the case of understanding gravitational waves they are especially suited to the task when a suitable tetrad is chosen. In both cases, we can understand the asymptotically flat vacuum solutions as modelling the space around an isolated source at that when static has mass m, in agreement with the Schwarzchild solution. The mass loss formula (2.7) is physically realisable most easily in a situation where the system is only non-static for a finite period $u \in (u_0, u_1)$. In this case, the mass of the source measured before and after will have a difference given by (2.7). In this way the Bondi mass is the analogue of mass for the non-static case. All information about the time-evolution of the radiation from the source in the form of gravitational waves is contained in the single complex news function (or equivalently in the real coefficient $c_{\mathcal{AB}}$ as discussed in 2.3). The important parameter of the Bondi shear, σ^0, as described in the spin coefficients is related to the asymptotic shear that is given in (2.6) in Section 2.3. We have that $M^a = rm^a$ so that (2.6) becomes

$$\sigma^0 = \lim_{r\to\infty} \left(r^2 m^a m^b \nabla_a \nabla_b u\right) = \lim_{r\to\infty} \left(r^2 \sigma\right) ,$$

where the limit and coefficient of r^2 being the area parameter does not affect the result (true as $r = \tilde{r} + O(\tilde{r})$ [12]), and the two quantities are thus equivalent. The coordinate system described in Section 2 is used in both approaches with a difference in definition for the coordinate parametrizing the null geodesics. The affine parameter $\tilde{r}$ is related to the unit sphere area parameter r by a transformation

$$r = \tilde{r} - \frac{1}{2}\sigma^0 \bar{\sigma}^0 \tilde{r}^{-1} + O(\tilde{r}^{-2})$$

given by Hayward [12]. Let us use this to give the relationship between the mass aspect and modified mass aspect. To denote quantities given in terms of the affine coordinate basis, a tilde will be used, otherwise it is given in the area parameter coordinate basis. The mass aspect can be equivalently defined as

$$M = -\frac{1}{2}\lim_{r\to\infty}\left[V - r\right] = \frac{1}{2}\lim_{r\to\infty}\left[r\left(\frac{g^{rr}}{g^{ru}} + 1\right)\right] ,$$

and the transformation of the components of the metric tensor can be worked out simply

$$g_{\boldsymbol{ab}} = \frac{\partial x^{\boldsymbol{a}}}{\partial x^{\tilde{\boldsymbol{a}}}} \frac{\partial x^{\boldsymbol{b}}}{\partial x^{\tilde{\boldsymbol{b}}}} g_{\tilde{\boldsymbol{a}}\tilde{\boldsymbol{b}}}$$

giving

$$\begin{aligned} g^{rr} &= g^{\tilde{r}\tilde{r}} \left(1 + \frac{1}{2}\sigma^0\bar{\sigma}^0\tilde{r}^{-2}\right)^2 + g^{\tilde{r}u}(\sigma^0\bar{\sigma}^0)_u\tilde{r}^{-1} + O(\tilde{r}^{-2})\,, \\ g^{ru} &= g^{\tilde{r}u}\left(1 + \frac{1}{2}\sigma^0\bar{\sigma}^0\tilde{r}^{-2}\right) + O(\tilde{r}^{-3})\,, \end{aligned}$$

where $g^{\tilde{r}\tilde{r}} = 2(U - \omega\bar{\omega})$, and $g^{\tilde{r}u} = 1$. Putting in the known quantities, we find

$$M = -\frac{1}{2}\left(\mathcal{M} + \bar{\mathcal{M}} - \bar{\eth}^2\sigma^0 - \eth^2\bar{\sigma}^0\right)\,,$$

and using the fact that $\mathcal{M} = \bar{\mathcal{M}}$, we find the relationship between mass aspect and modified mass aspect

$$\mathcal{M} = -M - Re\left(\eth^2\bar{\sigma}^0\right)\,,$$

and we can see that

$$-\oint \dot{\mathcal{M}}\, dS = \oint \dot{M}\, dS = \dot{m}(u)$$

via the $Re\,(\eth^2\dot{\bar{\sigma}}^0)$ integrating out, in correspondence with the term in (2.3).

Chapter 5

Conclusion

To conclude, we built up the Bondi-Sachs, and Newman-Penrose formalisms that are well suited to modelling gravitational waves, and put in the assumptions of vacuum solutions and asymptotic flatness to find asymptotic solutions for the Einstein equations. The kind of initial data and boundary conditions required for unique solutions was considered and given interpretation for both formalisms, and the connection between them was appreciated. We explored, in particular, the definition of Bondi mass and the Bondi mass loss and how these can be understood in either formalism, and the relationship to the initial data and boundary conditions. The way in which the Bondi mass loss relates to the understanding of gravitational wave energy was developed. In this sense understanding of these formalisms is important to understand how we can define energy and angular momentum of gravitational waves and can be used to make sense of physical observations. For a real world source, the asymptotically flat vacuum spacetime can be a good approximation to the behaviour of the outgoing waves far from the source.

In the modern day, asymptotic flatness is often understood in terms of a *conformal compactification* that introduces an unphysical spacetime where the $r \to \infty$ boundary is brought to a finite region [13]. When being used to understand gravitational waves, this is often cast in the language of the spin coefficients and was heralded by Penrose [14]. In order to understand, in general, the connection between definitions in terms of conformal compactification techniques and those of a asymptotic expansion in powers of the coordinates more work needs to be done. This could have applications in other fields where asymptotic expansion is commonly used, as if conformal compactification could be suitably related to the asymptotic expansion regime, the powerful theoretical machinery of conformal compactifications could be more easily harnessed. In this case, the coefficients of the $1/r$ expansion should be related in a systematic way to the quantities defined on null infinity in the conformal regime.

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Appendix A

Spinors

We can define the 2-spinors as arising from an algebra defined on *modules* over a commutative ring with unity for the specific case of complex fields.

A.1 Abstract index notation

Adopting the abstract index approach as in Penrose and Rindler [5], we start with a module $\mathbf{G}$ defined over a commutative ring with unity, $\mathcal{G}$, that has infinite copies distinguished by labels[1] $\{a, b, \dots\}$ called *indices*. Elements in the module $\mathcal{G}^a$ are denoted U^a, V^a, ..., and correspond to elements U^b, V^b, ..., respectively in the module $\mathcal{G}^b$, and are isomorphic to $\mathbf{G}$. We can then allow an operation of *index substitution* as a map $\mathcal{G}^a \to \mathcal{G}^b$, where we assign to an element V^a the corresponding element V^b. We may use this to replace the indices in any valid equation with another and get a different valid equation. The *duals*[2] of these modules are denoted by $\mathcal{G}_a$, $\mathcal{G}_b$ with the index down, and similarly have elements U_a, U_a. This allows us to define an operation, the *scalar product*, as a map $\mathcal{G}^a \times \mathcal{G}_a \to \mathcal{G}$, that takes the action of, say, $U_a \in \mathcal{G}_a$ on $V^a \in \mathcal{G}^a$, denoted by $U_a(V^a)$, $U_a V^a$ or $V^a U_a$. Note that as each of the $\mathcal{G}_a, \mathcal{G}_b \dots$, are necessarily isomorphic, we have $U_a V^a = U_b V^b = \cdots$. We also define elements of $\mathcal{G}$ to have valence $\left[\begin{smallmatrix}0\\0\end{smallmatrix}\right]$, while elements of the index-up modules and their duals have valences $\left[\begin{smallmatrix}1\\0\end{smallmatrix}\right]$ and $\left[\begin{smallmatrix}0\\1\end{smallmatrix}\right]$ respectively. Next, we can define the sets $\mathcal{G}^{a\dots r}_{g\dots b}$ (on disjoint subsets of the labels $\{a \dots r\}, \{g \dots b\}$) of arbitrary valence $\left[\begin{smallmatrix}p\\q\end{smallmatrix}\right]$ as the collections of $\mathcal{G}$-multilinear maps: $\mathcal{G}_a \times \cdots \times \mathcal{G}_r \times \mathcal{G}^g \times \cdots \times \mathcal{G}^b \to \mathcal{G}$, with elements denoted by $T^{a\dots r}_{g\dots b}$. This definition coincides with the given definitions if we assume that the module is reflexive, that is to say that the dual of $\mathcal{G}_a$ is isomorphic to $\mathcal{G}^a$.

On these elements we can define four basic operations. *Addition* is a mapping $\mathcal{G}^{a\dots r}_{g\dots b} \times \mathcal{G}^{a\dots r}_{g\dots b} \to \mathcal{G}^{a\dots r}_{g\dots b}$ that assigns to two elements $A^{a\dots r}_{g\dots b}$ and $B^{a\dots r}_{g\dots b}$ the multilinear map $\left(A^{a\dots r}_{g\dots b} + B^{a\dots r}_{g\dots b}\right)$ whose value is the sum for each input. *Outer multiplication* is a mapping $\mathcal{G}^{a\dots r}_{g\dots b} \times \mathcal{G}^{\alpha\dots\delta}_{\mu\dots\gamma} \to \mathcal{G}^{a\dots r\alpha\dots\delta}_{g\dots b\mu\dots\gamma}$, where each subset of labels is disjoint, that assigns to elements $A^{a\dots r}_{g\dots b}$ and $B^{\alpha\dots\delta}_{\mu\dots\gamma}$ an element $A^{a\dots r}_{g\dots b} B^{\alpha\dots\delta}_{\mu\dots\gamma}$ whose value is defined by the respective inputs on each element. This also happens to define scalar multiplication, if outer multiplication is taken with one of the

[1] Alternatively, we can think of the different modules as defined from elements of $\mathbf{G} \times \{a, b, \dots\}$ as pairs $\{G, a\}$ etc.

[2] That is, the modules defined over $\mathcal{G}$ as the collection of all linear mappings from $\mathcal{G}^a$ into $\mathcal{G}$ with scalar multiplication and addition defined to be linear in $\mathcal{G}^a$.

sets being $\mathcal{G}$. We can then see that the sets $\mathcal{G}^{a\ldots r}_{g\ldots b}$ are $\mathcal{G}$-modules with these operations, with some additional structure. Index substitutions can be performed in analogy with elements of $\mathcal{G}^a$ for elements of the general $\mathcal{G}^{a\ldots r}_{g\ldots b}$. Index substitution is a map from $\mathcal{G}^{a\ldots r}_{g\ldots b} \to \mathcal{G}^{\alpha\ldots\delta}_{\mu\ldots\gamma}$, where we make some change in the labelling set $\{a,\ldots r,g,\ldots,b\}$ to $\{\alpha,\ldots,\delta,\mu,\ldots,\gamma\}$, where the two sets may be overlapping and the upper-indices must each be equal in number with the lower-indices; in accordance with the indices changed we assign the corresponding element to $\mathcal{G}^{a\ldots r}_{g\ldots b}$ from $\mathcal{G}^{\alpha\ldots\delta}_{\mu\ldots\gamma}$. Again we may change any indices throughout an entire equation, and get a different valid equation.

In preparation to define the final basic operation we will henceforth adopt the approach that bold face indices stand for $1,\ldots,n$ in the regular way and apply the Einstein summation convention to these. Let us suppose that there exists a finite basis for $\mathcal{G}^a$; that is there exists some set $\delta^a_{\boldsymbol{a}} \in \mathcal{G}^a$, such that we have a unique expression $V^a = V^{\boldsymbol{a}}\delta^a_{\boldsymbol{a}}$, for $V^a \in \mathcal{G}^a$, and the elements $V^1, V^2, \cdots \in \mathcal{G}$ are the *components* in the given basis. It is then possible to show any other basis is constrained to have n elements, see Penrose and Rindler [5], pg 93. Then we take the *duals* of these elements to be the elements $\delta^{\boldsymbol{a}}_a \in \mathcal{G}^a$ that assign $\delta^{\boldsymbol{a}}_a V^a = V^{\boldsymbol{a}}$; these elements form a basis for $\mathcal{G}_a$ which can be easily verified via the product $\tilde{Q}_a \coloneqq (Q_b\delta^b_{\boldsymbol{b}})\delta^{\boldsymbol{b}}_a$ and its action on some $V^a \in \mathcal{G}^a$, for arbitrary $Q_b \in \mathcal{G}_b$. This basis is known as the *dual basis* to the original basis, and will almost exclusively be used as the basis for $\mathcal{G}_a$ given a basis for $\mathcal{G}^a$. We then define the components for Q_a with respect to the (dual) basis as $Q_{\boldsymbol{a}} = Q_a\delta^a_{\boldsymbol{a}}$. To express the components of a general element of $\mathcal{G}^{a\ldots r}_{g\ldots b}$ with respect to a given basis we define $A^{\boldsymbol{a}\ldots\boldsymbol{r}}_{\boldsymbol{g}\ldots\boldsymbol{b}} = A^{a\ldots r}_{g\ldots b}\delta^{\boldsymbol{a}}_a\ldots\delta^{\boldsymbol{r}}_r\delta^g_{\boldsymbol{g}}\ldots\delta^b_{\boldsymbol{b}}$. It then follows that $A^{a\ldots r}_{g\ldots b} = A^{\boldsymbol{a}\ldots\boldsymbol{r}}_{\boldsymbol{g}\ldots\boldsymbol{b}}\delta^a_{\boldsymbol{a}}\ldots\delta^r_{\boldsymbol{r}}\delta^{\boldsymbol{g}}_g\ldots\delta^{\boldsymbol{b}}_b$.

We can now define *contraction*, which is easiest to define once we have a conception of a basis[3]. The $\binom{\mu}{\gamma}$-contraction operation will be defined as a map $\mathcal{G}^{a\ldots r\mu}_{g\ldots b\gamma} \to \mathcal{G}^{a\ldots r}_{g\ldots b}$, where neither μ nor γ belong to either of the disjoint sets $\{a,\ldots r\},\{g,\ldots,b\}$, that is defined for an element $T^{a\ldots r\mu}_{g\ldots b\gamma}$, under some basis $\delta_{\boldsymbol{a}}{}^a$, as follows:

$$T^{a\ldots r\mu}_{g\ldots b\gamma} = T^{\boldsymbol{a}\ldots\boldsymbol{r}\boldsymbol{\mu}}_{\boldsymbol{g}\ldots\boldsymbol{b}\boldsymbol{\gamma}}\delta^a_{\boldsymbol{a}}\ldots\delta^r_{\boldsymbol{r}}\delta^{\boldsymbol{g}}_g\ldots\delta^{\boldsymbol{b}}_b\delta^{\boldsymbol{\mu}}_{\boldsymbol{\mu}}\delta^{\boldsymbol{\gamma}}_{\gamma} \to T^{\boldsymbol{a}\ldots\boldsymbol{r}\boldsymbol{\mu}}_{\boldsymbol{g}\ldots\boldsymbol{b}\boldsymbol{\gamma}}\delta^a_{\boldsymbol{a}}\ldots\delta^r_{\boldsymbol{r}}\delta^{\boldsymbol{g}}_g\ldots\delta^{\boldsymbol{b}}_b\delta^{\boldsymbol{\mu}}_{\boldsymbol{\mu}}\delta^{\boldsymbol{\gamma}}_{\mu}\,,$$

denoted by $T^{a\ldots r\mu}_{g\ldots b\mu}$, and this will define a new tensor that is independent of basis. Noting that $T^{a\ldots r\mu}_{g\ldots b\mu} = T^{a\ldots r\gamma}_{g\ldots b\gamma} = T^{a\ldots r\alpha}_{g\ldots b\alpha}$ via the construction of the scalar product, leading to the designation of the two contracted indices as *dummy indices* that can be changed at will so long as they don't belong to either of the index sets.

All this being said, the abstract elements essentially behave the same as familiar components for all the algebraic properties - the difference is almost entirely conceptual. However, once covariant differentiation is introduced the abstract and component indices will behave quite differently.

A.2 Spin-vectors

In order to discuss the abstract formulation of spinors it will be helpful to have an understanding of *spin-vectors* on a space-time manifold. Let us establish Minkowski space with

[3]It is in fact, not possible to define a contraction operation on the general elements as defined, we require the the modules $\mathcal{G}^a$ to be *totally reflexive*, for which a finite basis is a sufficient but not necessary condition. See Penrose and Rindler [5], pg 84.

flat metric as $\mathbb{M}$ with convention $\eta_{ab} = \text{diag}\,(1,-1,-1,-1)$. We would like to consider expressions of vectors in $\mathbb{M}$ primarily in terms of restricted minkowski tetrads, $\boldsymbol{e_a}$, that is bases that are orthochronous, proper, and satisfy $\boldsymbol{e_a}\cdot\boldsymbol{e_b} = \eta_{\boldsymbol{ab}}$. We will denote the components of a vector, $\boldsymbol{U}$, as $U = T\boldsymbol{t} + X\boldsymbol{x} + Y\boldsymbol{y} + Z\boldsymbol{z}$ for tetrad $(\boldsymbol{t},\boldsymbol{x},\boldsymbol{y},\boldsymbol{z})$. Let us establish the idea of a null cone, that is the set of all null vectors. Timelike or null vectors $\boldsymbol{U}$ live inside this cone. Now consider the $T = 0$ hyperplane to split the null cone into two distinct halves, which are the future null cone and past null cones for those above and below the origin respectively. If we wish to consider only the directions (considering $\pm\boldsymbol{U}$ to have opposite and unequal direction) of the null vectors, we could represent those directions by the coordinates of the intersection points of the future or past null cones with $T = \text{const}$ hyperplanes (for $T \neq 0$). Let us take $T = \pm 1$ for the future and past cones respectively for simplicity. As, of course, we can represent any direction with the intersection of the vector and one of the hyperplanes, the intersections can be understood as two separate 2-spheres that have interiors representing timelike directions, and exteriors representing spacelike directions for the future and past null cones. As the future and past null cones have directions that are associated with one another, we can consider these spheres to be the opposite of one another in the sense that the coordinates have inverted sign for the corresponding opposite sphere. Taking the future sphere, to be situated in the complex plane as a Riemann sphere, we can associate each point on it with a number in the complex plane, provided we attach infinity which will capture the north pole. If the sphere is defined as $x^2+y^2+z^2 = 1$, lowercase letters denoting coordinates on the Riemann sphere, then we map $(1,x,y,z) \to (1,X,Y,0)$ and label this mapped point with the single complex parameter $\xi = X + iY$ and find that

$$\xi = \frac{x+iy}{1-z}\,,$$

where we can make the same construction analogously for the past sphere. If we now re-state ξ in terms of two complex numbers ζ,η as $\xi = \frac{\zeta}{\eta}$, we can avoid the use of an infinite coordinate. We note at this stage that taking complex linear transformations of these components, where the matrix representing these

$$\zeta' = a\zeta + b\eta,\ \eta' = c\zeta + d\eta \leftrightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is non-singular, form the operation for a group over the set of these components. Without loss of generality, we can take $ad - bc = 1$ and then the operations are referred to as spin transformations. We have then that each pair (ζ,η) represents a null vector, $\boldsymbol{K}$, that is known as its *flagpole*. We find that there is a redundancy in the pair (ζ,η) in representing a null vector up to a phase transformation, $\zeta \to e^{i\theta}\zeta, \eta \to e^{i\theta}\eta$ for θ real. Let us now introduce a real spacelike vector L, that is both tangent to the Riemann sphere, and orthogonal to K. This vector defines a 2-plane together with K given by $\text{s}L + \text{v}K$ for $s,v \in \mathbb{R}$, or a half plane if v is taken to be $v > 0$. This half plane is known as a *flag plane*. Together with the flagpole they are known as the *null flag*. The flag plane and flagpole together determine (ζ,η) completely up to a sign ambiguity. To see this let us take the transformation where θ varies from 0 to π, this changes the pair into $(-\zeta,-\eta)$ but rotates the flag plane completely through 2π, and is thus an equivalent representation of the null flag. Considering the rotation of π as corresponding to a full rotation 2π in the Minkowski space, this introduces the idea

of *spinoral objects* that are not sent to their original state after a rotation of 2π about an axis is performed, but instead are after a rotation of 4π. We define spin vectors as elements of *spin space*, defined itself as a complex vector space in which there exists an inner product on a pair of spin-vectors κ, and ω, denoted by $\{\kappa, \omega\}$ that is skew-symmetric, bilinear and non-degenerate. We can associate the elements of such a space with the null flags, where each null flag defines two spin-vectors κ and $-\kappa$, with the pair of numbers (ζ, η) representing the components of κ. Taking some reference system as given, in this case we define the inner product to function with the components in the following way, if (ζ_0, η_0) represents a spin-vector κ and (ζ_1, η_1) represents a spin-vector ω, then $\{\kappa, \omega\} = \{(\zeta^0, \eta^0), (\zeta^1, \eta^1)\} = \zeta^0\eta^1 - \eta^0\zeta^1$, while addition and scalar multiplication are defined in the obvious ways.

A.3 2-Spinor algebra

If we apply the above construction in Section A.1 to the module consisting of tangent vector spaces ($\mathcal{G} = \mathbb{R}$) or tangent vector fields ($\mathcal{G} = C^\infty$ real scalar fields) on a space-time manifold, we produce the standard tensor algebra employed in general relativity with the benefits of the abstract index construction. If we instead apply it to the spin-vectors at a point ($\mathcal{G} = \mathbb{C}$) or spin-vector fields ($\mathcal{G} = C^\infty$ complex scalar fields) on a space-time manifold we produce the spinor algebra. The elements of general modules $\mathcal{G}^{A\cdots R}_{G\cdots B}$ are defined as *spinors*. We will use the notation that capital indices are to be used when denoting spinors. The only addition we must make to the basic theory is to involve the anti-symmetrical inner product that is defined for spin-vectors. The anti-symmetrical inner product is bilinear in $\mathcal{G}$ so we can define a spinor $\varepsilon_{AB} \in \mathcal{G}_{AB}$ that is uniquely defined from this action. This plays the role of the metric tensor as in standard tensor theory but is anti-symmetric $\varepsilon_{AB} = -\varepsilon_{BA}$. We use it to establish a mapping between the modules $\mathcal{G}^A$, $\mathcal{G}^B$, ..., and their duals, $\mathcal{G}_A$, $\mathcal{G}_B$, ..., that is; for some $\kappa^A \in \mathcal{G}^A$ we map to the associated dual element $\kappa_A \in \mathcal{G}_A$ via $\kappa^A \to \kappa_B = \kappa^A\varepsilon_{AB}$ using the same letter to denote this connection between the elements. From the properties of this mapping as a spinor representation of the spin-vector inner product, we see that it is an isomorphism and thus there exists an inverse mapping, that defines an element $\varepsilon^{AB} \in \mathcal{G}^{AB}$ that assigns $\kappa_A \in \mathcal{G}_A$ to an element $\kappa^A \in \mathcal{G}^A$ via $\kappa_A \to \kappa^A = \kappa_B\varepsilon^{AB}$. These mappings are inverses in the sense that $\varepsilon_{AB}\varepsilon^{BC} = \varepsilon_A{}^C = \delta^C_A$, where the symbol δ^C_A represents the defined isomorphism between sets $\mathcal{G}^A$ and $\mathcal{G}^C$, and we notice that the notation is consistent if we regard the epsilon operators to act on each other to raise or lower an index. Because we may now raise or lower any index, it is important to stagger the indices so that we have a position for which any index may move to. As the ϵ spinor is defined through the inner product for spin-vectors we may now write $\{\kappa, \omega\} = \kappa_A\omega^A$ and, by the definitions in Appendix A.2, we have that

$$\kappa_A\omega^A\tau^B + \omega_A\tau^A\kappa^B + \tau_A\kappa^A\omega^B = 0\,,$$

which, through the arbitrary choice of spin-vectors $(\kappa^A, \omega^A, \tau^A)$ and appropriate lowering, implies

$$\varepsilon_{AB}\varepsilon_{CD} + \varepsilon_{BC}\varepsilon_{AD} + \varepsilon_{CA}\varepsilon_{BD} = 0\,. \tag{A.1}$$

This is an identity we will use later. Now the above defines a system of spinor algebra that is self-contained but, may we wish to relate these spinors to familiar tensors, is insufficient. To

meet this end, the final base operation that completes our 2-spinor algebra will be defined: *complex conjugation*. Let us define a different group of sets, with their own labels, denoted with a prime. These read $\mathcal{G}^{A'}$, $\mathcal{G}^{B'}$, ..., and are related to the original $\mathcal{G}^{A}$ by a mapping $\mathcal{G}^{A} \to \mathcal{G}^{A'}$: $\kappa^{A} \to \overline{\kappa^{A}} = \bar{\kappa}^{A'}$. This mapping defines addition and scalar multiplication in these new sets are by imposing

$$\nu\kappa^{A} + \sigma\omega^{A} = \tau^{A} \iff \bar{\nu}\bar{\kappa}^{A'} + \bar{\sigma}\bar{\omega}^{A'} = \bar{\tau}^{A'},$$

where $\nu, \sigma \in \mathcal{G}$, $\kappa^{A}, \omega^{A}, \tau^{A} \in \mathcal{G}^{A}$, and we utilize the familiar complex conjugation on elements of $\mathcal{G}$. Then the new sets are also $\mathcal{G}$-modules, isomorphic to the original modules[4], related via the operation of complex conjugation. The inverse map is denoted with the same symbol, and we note $\overline{\bar{\kappa}^{A'}} = \kappa^{A}$ for any $\kappa^{A} \in \mathcal{G}^{A}$. We can then define dual modules from the primed modules, which have a similar mapping with the duals of the original modules.

The *general 2-spinor* (hereafter: simply a spinor) is thus defined as a $\mathcal{G}$-multilinear map $\mathcal{G}_{A} \times \cdots \times \mathcal{G}_{R} \times \mathcal{G}_{A'} \times \cdots \times \mathcal{G}_{R'} \times \mathcal{G}^{G} \times \cdots \times \mathcal{G}^{B} \times \mathcal{G}^{G'} \times \cdots \times \mathcal{G}^{B'} \to \mathcal{G}$, where each of the index sets are disjoint, considering A, A' to be separate. On these spinors, as elements of the sets $\mathcal{G}^{A\ldots RA'\ldots R'}_{G\ldots BG'\ldots B'}$, we can define the four basic operations of outer multiplication, addition, index substitution and contraction acting individually on each of the primed and unprimed indices as in Section A.1. We now can have another operation of complex conjugation on spinors that takes the complex conjugate of the result of the map, defining a mapping in the complex conjugate space of each of the $\mathcal{G}_{A}, \ldots, \mathcal{G}^{B'}$ sets. Together with the complex conjugate of ε_{AB} denoted as $\varepsilon_{A'B'}$ (without the bar), that are allowed to raise and lower elements in the primed space via an induced isomorphism from ε_{AB}, the 2-spinor algebra is set up.

There are a few things left to discuss. First, note that the operations of addition and scalar multiplication, in the special case where one of the sets is $\mathcal{G}$, give all the general $\mathcal{G}^{A\ldots RA'\ldots R'}_{G\ldots BG'\ldots B'}$ sets the structure of $\mathcal{G}$-modules. Next, index substitutions can only act on separate permutations of the primed indices and unprimed indices so that the order of these in a representation of a spinor is unimportant. Also, we find that the contraction operation for both primed and unprimed indices commutes with addition, outer multiplication, and index substitution, while complex conjugation commutes with all four basic operations as one would expect. Finally we discuss bases; the discussion in Section A.1 can be applied to spinors where we denote a general basis of $\mathcal{G}^{A}$ as $\varepsilon_{\boldsymbol{A}}{}^{A}$, known as a *dyad*, owing to the two-dimensional nature of a basis for $\mathcal{G}^{A}$. Given a basis for $\mathcal{G}^{A}$ there is a natural basis for $\mathcal{G}^{A'}$ of the complex conjugates of basis elements given, $\varepsilon_{\boldsymbol{A'}}{}^{A'}$, this is what we shall default to. In the particular case that the basis is a pair of spinors, $\varepsilon_{\boldsymbol{A}}{}^{A} = (o, \iota)$, where $o_{A}\iota^{A} = 1$, then it is known as a *spin-frame* and has special properties. In the case of a spin-frame we make the convention that the induced basis elements for $\mathcal{G}^{A'}$ are denoted $(o^{A'}, \iota^{A'})$, without the bars, for ease of writing. A spin-frame of course induces a dual basis which we find has elements $\varepsilon_{A}{}^{\boldsymbol{A}} = (-\iota_{A}, o_{A})$ which means that components of an arbitrary $\kappa^{A} \in \mathcal{G}^{A}$ and κ_{A} are related via $\kappa_{0} = \kappa^{1}, \kappa_{1} = \kappa^{0}$. We can also, in the case of a spin-frame write $\varepsilon^{AB} = o^{A}\iota^{B} - \iota^{A}o^{B}$.

[4]It turns out these sets are actually *anti-isomorphic*.

A.4 Tensors as embedded in 2-spinors

We define a tensor labelling set from clumped pairs of spinor indices, one prime and one unprimed, denoted as $\{a, b, \dots\}$, where $a = AA'$, $b = BB'$ etc. Then we can embed an algebra in the algebra of 2-spinors, (*world-tensor* algebra) by only working with elements of the $\mathcal{G}, \mathcal{G}^a, \mathcal{G}^b_{zx}, \dots$ sets. These sets are special under complex conjugation as they are sent to themselves, and so an elements of them under complex conjugation produce another element within the same set. These elements will be called *complex world-tensors* and have all the operations as in Section A.3 defined on them, while being closed under these operations. Restricting ourselves again to elements that have the property of invariance under complex conjugation we have, under the operations defined in Section A.3, another closed system of spinors known as *real world-tensors*. We will denote the subsets of the world tensor modules that only consist of real elements as $\mathcal{N}, \mathcal{N}^a, \mathcal{N}^b_{zx}, \dots$, respectively. The tensor system generated under real world-tensors has $\mathcal{N}$ as the real numbers at a point, or real scalar fields over the space-time manifold in A.3. If we define the real world tensor $g_{ab} = \varepsilon_{AB}\varepsilon_{A'B'}$, then the algebra is identical to the standard tensor algebra employed in general relativity, with g_{ab} as the standard metric tensor, so we will simply call these spinors tensors, or complex tensors when considering all of the complex world-tensor sets.

A.5 Decomposition of the Riemann tensor

We can decompose the Riemann tensor into spinor parts by the following analysis. Taking the Riemann tensor $R_{abcd} = R_{AA'BB'CC'DD'}$ in accordance with Appendix A.4, we have

$$R_{abcd} = R_{AB[A'B]'CD[C'D']} + R_{AB[A'B]'[CD]C'D'} + R_{[AB]A'B'CD[C'D']} + R_{[AB]A'B'[CD]C'D'} \,,$$

which we can find by applying the symmetries of the Riemann. Whenever a spinor is antisymmetric in two indices it is possible to perform a decomposition of the type

$$T_{[AB]} = \frac{1}{2} T_C{}^C \varepsilon_{AB} \,, \tag{A.2}$$

which can be seen by raising indices C, D of (A.1) and contracting by $T_{[CD]}$. Using (A.2) we have

$$\begin{aligned} R_{abcd} = {} & X_{ABCD}\varepsilon_{A'B'}\varepsilon_{C'D'} + \overline{X}_{A'B'C'D'}\varepsilon_{AB}\varepsilon_{CD} \\ & + \Phi_{ABC'D'}\varepsilon_{A'B'}\varepsilon_{CD} + \overline{\Phi}_{A'B'CD}\varepsilon_{AB}\varepsilon_{C'D'} \,, \end{aligned}$$

where $X_{ABCD} = \frac{1}{4} R_{ABE'}{}^{E'}{}_{CDF'}{}^{F'}$, and $\Phi_{ABC'D'} = \frac{1}{4} R_{ABE'}{}^{E'}{}_{F}{}^{F}{}_{C'D'}$. Straightforwardly translating the first and second pair antisymmetries of the Riemann tensor into these spinors, we can write $X_{ABCD} = X_{(AB)(CD)}$, $\Phi_{ABC'D'} = \Phi_{(AB)(C'D')}$. If we define $\Lambda = \frac{1}{6} X_{AB}{}^{AB}$, then the cyclic identity and pair interchange identities imply that $\Lambda = \bar{\Lambda}$. Finally we can decompose the spinor X_{ABCD} into totally symmetric parts using the established symmetries plus the pair interchange (which yields $X_{ABCD} = X_{CDAB}$) to find

$$X_{ABCD} = \Psi_{ABCD} + \Lambda \left(\varepsilon_{AC}\varepsilon_{BD} + \varepsilon_{AD}\varepsilon_{BC} \right) \,,$$

if $\Psi_{ABCD} \coloneqq X_{(ABCD)}$, the totally symmetric part.